

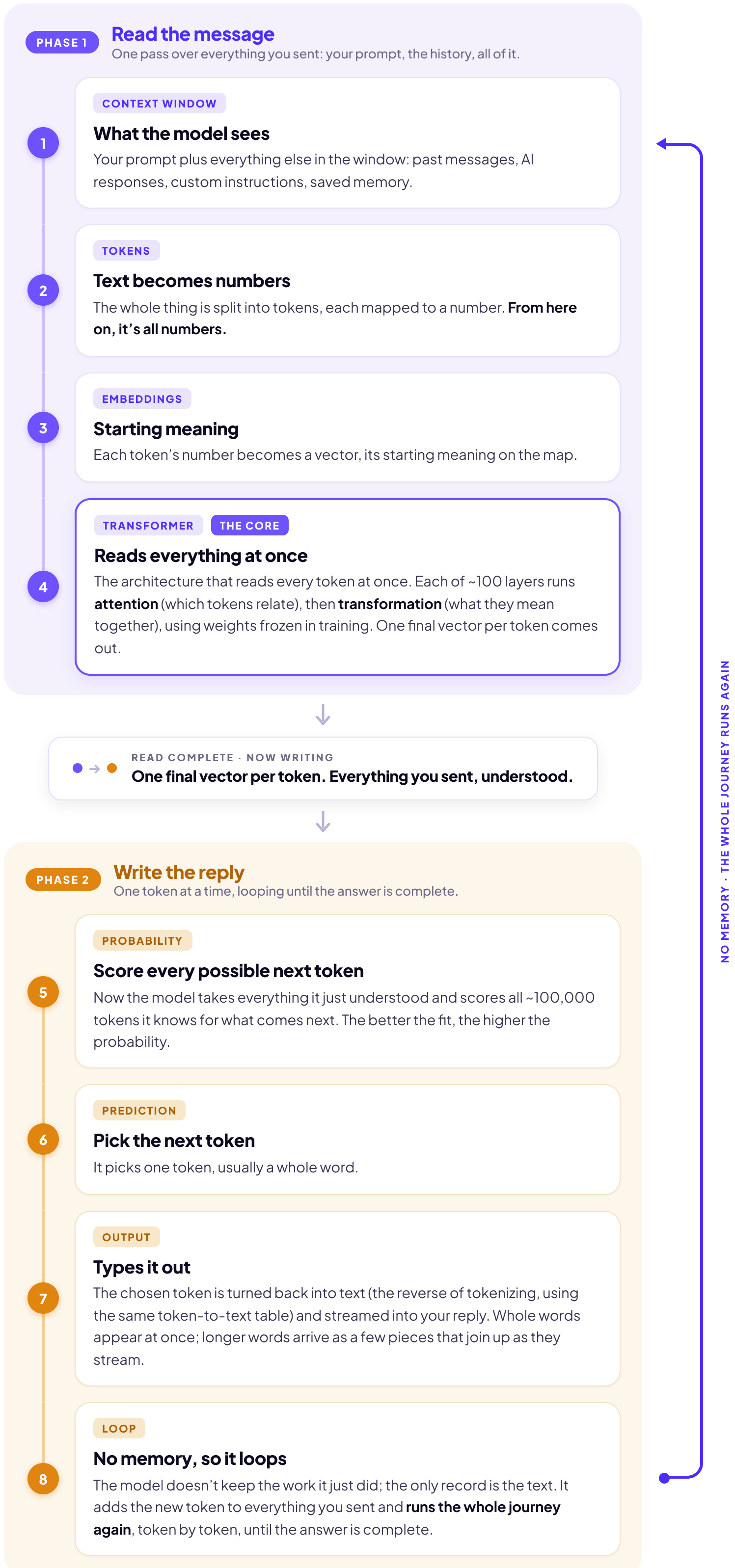
Inference

You've met every piece on its own: tokens, embeddings, layers, the context window, and the single prediction that picks one token. Now watch them work together as a single journey from your prompt to a finished answer.

Your prompt runs all the way through the model to make one prediction: a single token. Then it runs again for the next, and again, until the answer is complete. That whole repeating journey has a name: **Inference**.

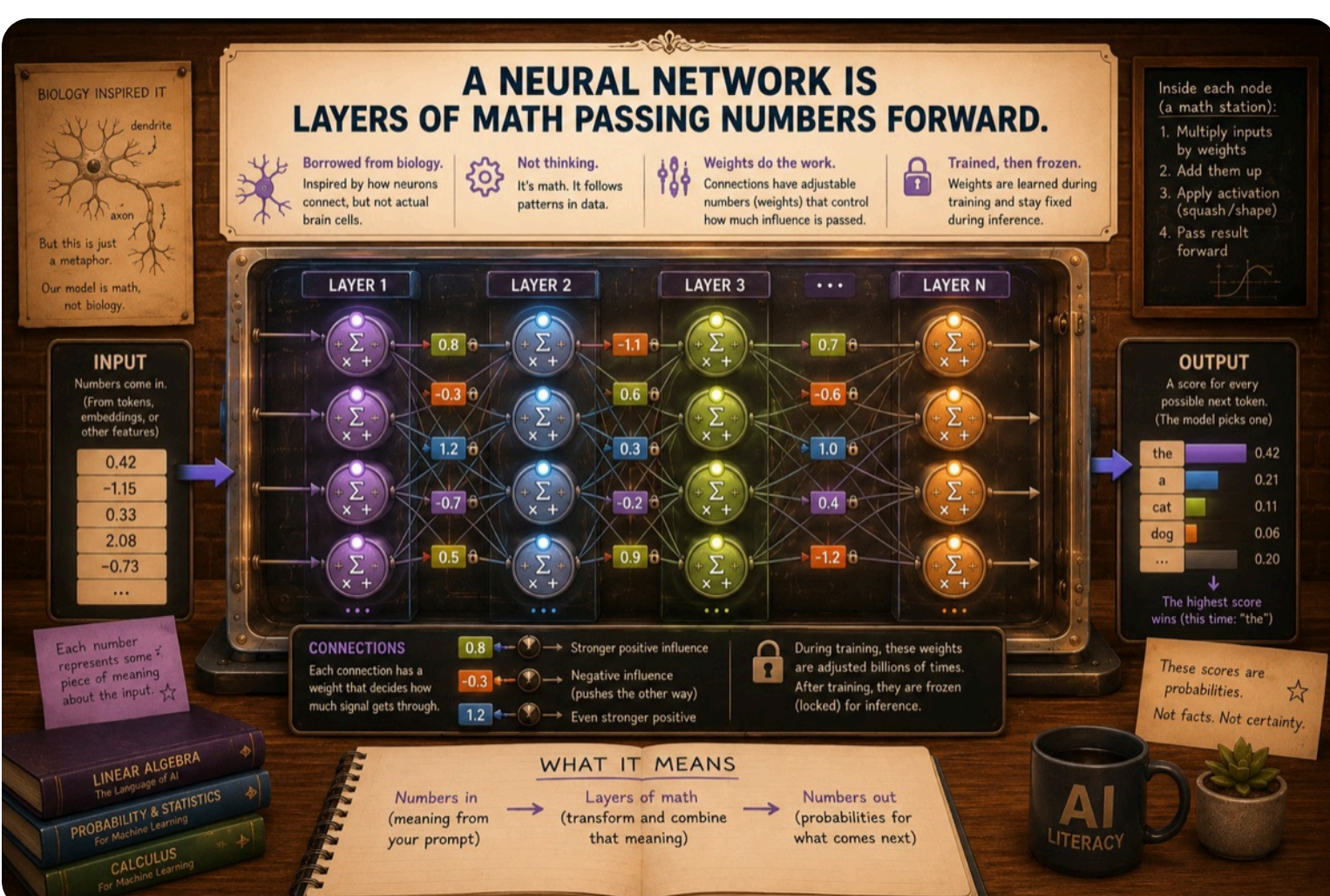
It isn't training again. It's using what training already built, reading those frozen patterns back out one token at a time.

Here's the whole journey in one picture, from your prompt to a finished answer. **It runs in two phases, and the end feeds back to the start.**



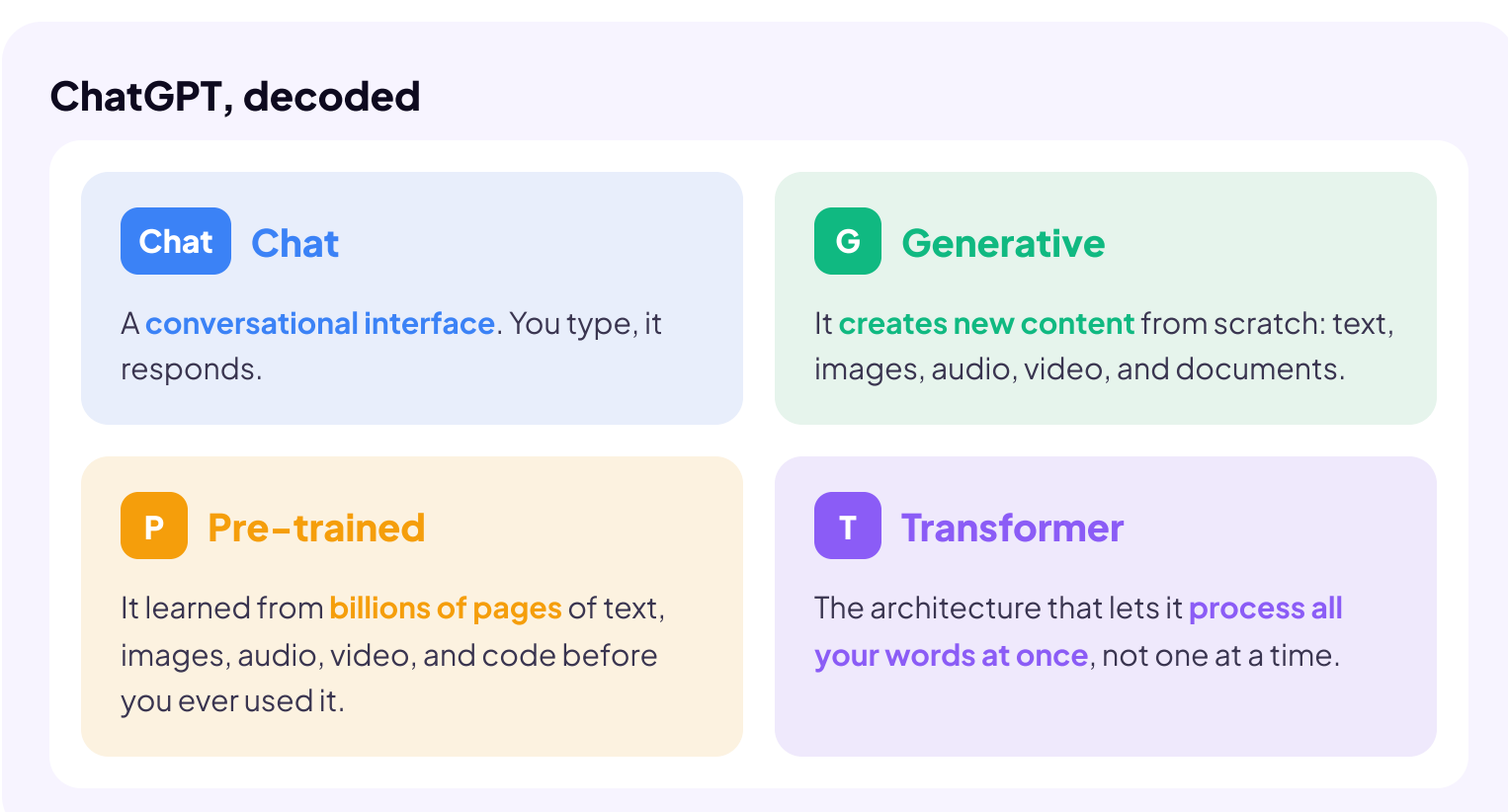
THE ENGINE, RUNNING FROZEN

At the center of that whole journey is the neural network you met in Layers: the stacked nodes and **weights**. Training set those weights; inference just runs them forward, the same machine doing the same arithmetic every time you ask.



READ THE NAME ON THE BOX

One last payoff before you move on. You've now seen every piece of how an LLM works, and here's the proof: you can read the most famous name in AI like a spec sheet. Every letter is something you just learned.



LOCK IN THE PIECES

You've now met every piece of how an LLM works: tokens and embeddings, attention and layers, prediction and training, the context window, and inference itself. Here's a quick checkpoint to lock them in.